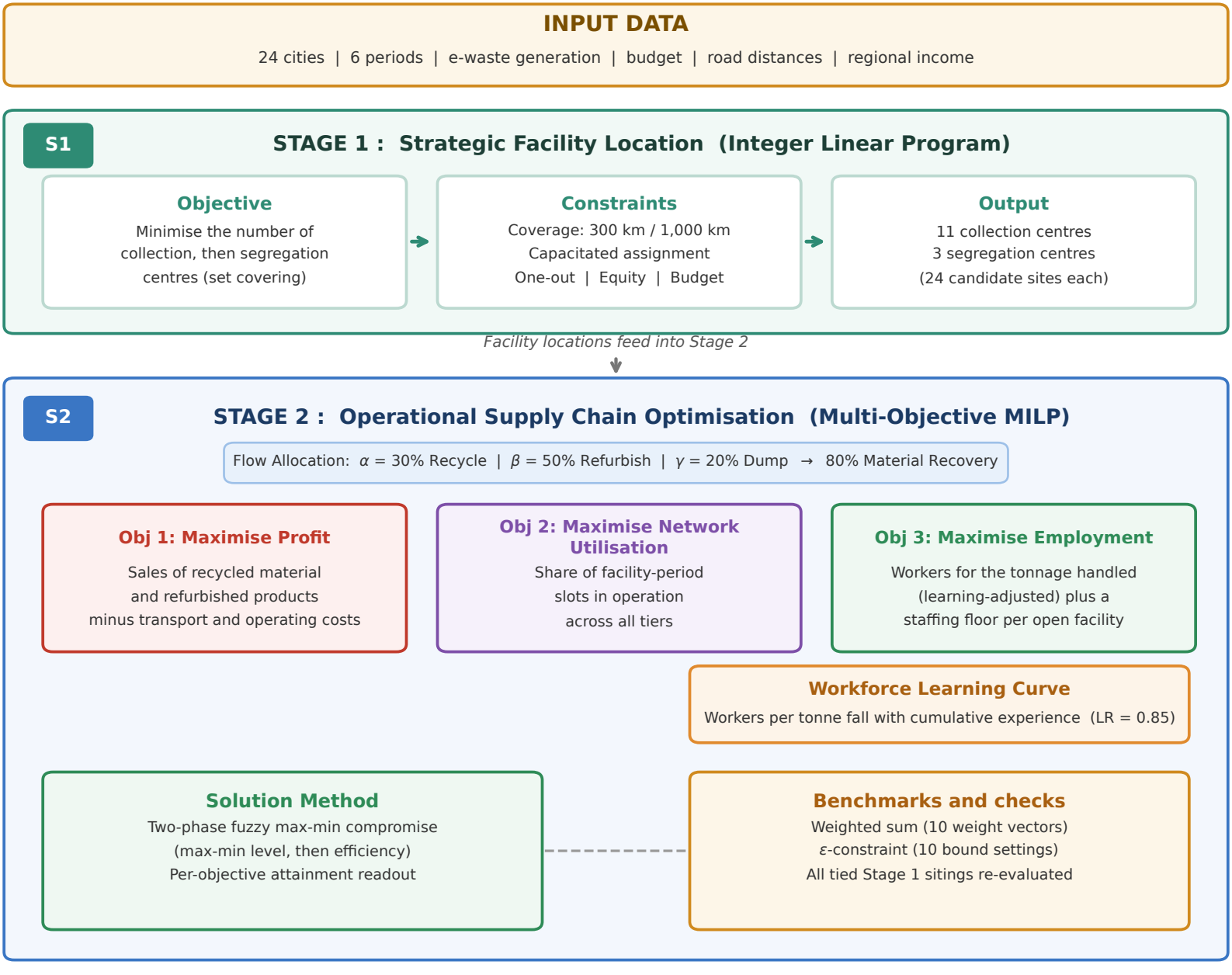
INPUT DATA
24 cities | 6 periods | e-waste generation | budget | road distances | regional income
S1
STAGE 1 : Strategic Facility Location (Integer Linear Program)
Objective
Minimise the number of collection, then segregation centres (set covering)
Constraints
Coverage: 300 km / 1,000 km
Capacitated assignment
One-out | Equity | Budget
Output
11 collection centres
3 segregation centres
(24 candidate sites each)
Facility locations feed into Stage 2
S2
STAGE 2 : Operational Supply Chain Optimisation (Multi-Objective MILP)
Flow Allocation: α = 30% Recycle | β = 50% Refurbish | γ = 20% Dump → 80% Material Recovery
Obj 1: Maximise Profit
Sales of recycled material and refurbished products minus transport and operating costs
Obj 2: Maximise Network Utilisation
Share of facility-period slots in operation across all tiers
Obj 3: Maximise Employment
Workers for the tonnage handled (learning-adjusted) plus a staffing floor per open facility
Workforce Learning Curve
Workers per tonne fall with cumulative experience (LR = 0.85)
Solution Method
Two-phase fuzzy max-min compromise
(max-min level, then efficiency)
Per-objective attainment readout
Benchmarks and checks
Weighted sum (10 weight vectors)
ε-constraint (10 bound settings)
All tied Stage 1 sitings re-evaluated